PROOFLINE

Evidence-Backed Brand Governance —
Verify Before You Publish

Presented By : Binary Brains





PROBLEM STATEMENT

Content creation scaled 10x with AI, but businesses still struggle to verify that every claim matches their latest approved facts.

WHERE TRADITIONAL REVIEW BREAKS

- **Disconnected Guidelines:** Brand rules sit in static PDFs that creators and AI tools never check while drafting.
- **Contradicted Claims:** Marketing copy frequently clashes with real specs (e.g., claiming 20h battery instead of approved 14h).
- **Expired Information:** Deprecated pricing and lapsed offers slip into live campaigns without version checks.
- **Bottlenecked Manual Review:** Reviewers waste hours repeatedly cross-checking docs instead of high-value work.



TARGET USERS

WHO USES THE WORKFLOW?

Different roles. One shared pre-publication quality gate

- **CONTENT CREATORS:** Catch contradicted specs and get instant, intent-preserving corrections before submission.
- **CAMPAIGN MANAGERS:** Triage flagged drafts by risk score, compare revisions, and track approval status.
- **BRAND REVIEWERS:** Inspect evidence-linked sources and log traceable approvals (no blind trust in AI).
- **FACT & BRAND OWNERS:** Maintain the single source of truth—versioning specs, validity dates, and brand rules.





THE SOLUTION

A Pre-Publication Quality Gate Powered by Source-Grounded AI

Brand Truth Registry + Claim Ledger

Centralises versioned brand specs and links every detected claim to its exact source document and validity date.

Deterministic & Grounded AI Auditing

Combines exact deterministic rules (numbers, pricing, banned terms) with source-grounded LLM context checks.

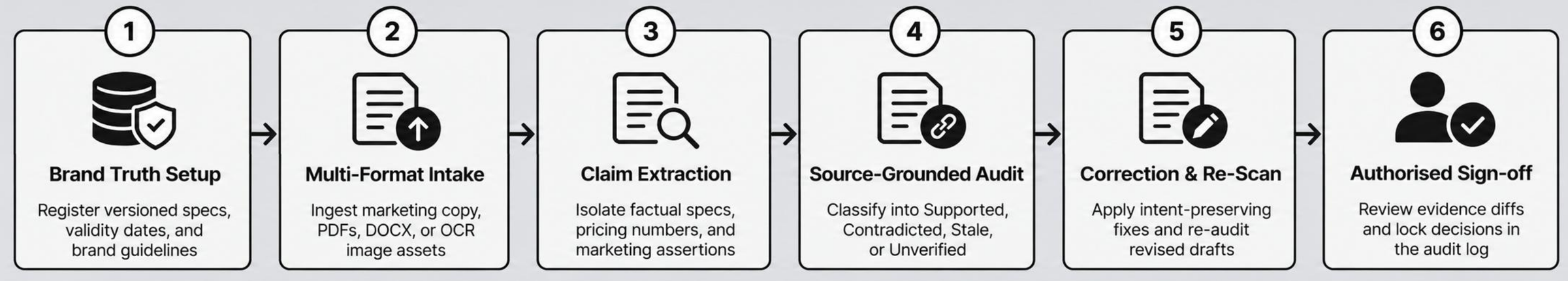
Evidence-or-Abstain Reliability

Detects expired promos and contradicted specs; marks missing evidence as 'Needs Human Review' instead of hallucinating.

Human-in-the-Loop Review & Audit Trail

Suggests intent-preserving fixes, supports re-auditing revised drafts, and locks final sign-off to authorised humans.

COMPLETE PRODUCT WORKFLOW



CORE WORKFLOW GUARANTEES:

- **Versioned Truth Registry:**
  Tracks exact source citations, effective dates, and fact expiration windows.
- **Hybrid Audit Engine:** Deterministic checks for numbers/dates + LLM checks for context and tone.
- **Evidence-or-Abstain:** Every flag cites an approved record; missing evidence flags 'Needs Review' instead of guessing.
- **Immutable Lineage:** Preserves original drafts, accepted edits, and side-by-side revision comparisons.
- **Zero Auto-Publishing:** Content is never published automatically; final authority stays with an authorised human.

FEATURES & TECHNOLOGY STACK

Implemented system architecture and verified build stack

CORE IMPLEMENTED CAPABILITIES

01

Brand Truth Registry

Stores versioned facts, source citations, validity windows, and brand guidelines.

02

Claim Grounding & Audit Engine

Extracts claims and audits them against sources: Supported, Contradicted, or Unverified.

03

Intent-Preserving Correction & Re-audit

Generates minimal diff-based fixes and re-audits revised drafts instantly.

04

Human-in-the-Loop Audit Trail

Enforces role-based sign-off and records an immutable, timestamped review history.

VERIFIED BUILD STACK

Frontend	Next.js (App Router), TypeScript, Tailwind CSS
Backend	Python 3.12, FastAPI, Pydantic v2
Database & Authentication	SQLite / D1 & Postgres schemas, scoped memory vault
AI Layer	Schema-enforced LLM adapter with strict JSON output validation
Verification Engine	Deterministic exact-match rules + source lexical retrieval
Document Processing	Local document extraction (PDF / DOCX) + bounded OCR parser

Evidence-or-Abstain
Missing evidence yields 'Needs Review', not hallucinations.

Deterministic-First
Numbers, pricing, and dates validated by code, not LLM guesses.

Human Accountability
Zero auto-publishing; strict separation of creator and approver.

RESPONSIBLE AI & DATA SECURITY

Trust, transparency, and human accountability

01

Tenant Isolation & Data Privacy

Enforces strict organisation workspace boundaries, keeps API secrets strictly server-side, and stores zero client keys.

02

Evidence-or-Abstain Engine

Grounds every finding in a versioned source citation; flags missing or conflicting evidence as 'Needs Review' instead of hallucinating.



03

Accountable Human Sign-off

Enforces role separation (creators cannot approve own work), logs override reasons, and strictly blocks auto-publishing.

04

Explicit Disclosure & Fail-Soft States

Dedicated /privacy disclosure route, transparent parsing-failure alerts, and clear disclaimers that Proofline is governance, not legal counsel.

OUR PRINCIPLE: AI assists verification. Humans remain accountable for final approval.

DEVELOPMENT PLAN & EXPECTED IMPACT

From a focused MVP to a reliable brand-governance platform

1 ROUND 1

PLAN & VALIDATE (COMPLETED)

- Define the problem, target users and MVP scope.
- Finalise workflow, architecture, and technology choices.
- Assign team responsibilities and validation tasks.

2 ROUND 2

CORE BUILD & BASELINES (DELIVERED)

- **Delivered:** Versioned Brand Truth Registry, validity dates, and multi-format intake.
- **Delivered:** Hybrid audit engine (deterministic spec rules + source-grounded findings).
- **Delivered:** Intent-preserving diff fixes, re-audit pipeline, and role-based sign-off.
- **Delivered Baselines:** Claim–fact lineage, explicit abstain states, and review metrics.

3 ROUND 3

EXPANDED CAPABILITIES (NEXT)

- Full Fact-Change Impact Radar: Automated re-audit queue when specs change.
- High-severity verifier pass and edge-case benchmark test corpus.
- Guideline-Gap Detector: Proposes rule clarifications for recurring mistakes.
- Structured user testing trials and live external provider integration.

EXPECTED IMPACT

Reduce Manual Effort
Eliminates repetitive manual spec lookups, accelerating draft-to-review velocity.

Identify Risks Early
Catches contradictory specs, expired promotions, and missing qualifiers pre-publication.

Improve Traceability
Provides an immutable audit log of source versions, diffs, and reviewer sign-offs.

FUTURE SCOPE

Platform Integrations
CMS connectors (WordPress, Webflow), Google Docs add-on, and Slack notifications.

Expanded Content Support
Multi-image visual asset matching, video transcripts, and multi-language audits.

Advanced Analytics
Cross-team compliance heatmaps and recurring mistake telemetry.

Validation Note: Impact targets will be measured in user trials; no fabricated production metrics claimed.

TEAM & RESPONSIBILITIES

Verified ownership & core engineering contributions



01

[Insert Name] — Lead & Architecture

Problem validation, PRD governance specs, workflow architecture, and submission delivery.

02

[Insert Name] — AI & Verification Engine

Deterministic rule engine, claim extraction, schema validation, and Evidence-or-Abstain pipeline.

03

[Insert Name] — Backend & Platform

FastAPI services, in-memory vault, D1/Postgres schemas, and role-based permissions.

04

[Insert Name] — Frontend & UX Engineering

Next.js App Router, responsive audit queue, diff-based correction editor, and review portal.

05

[Insert Name] — QA & Systems Transparency

E2E/API test suites, fail-soft error handling, AI/Privacy disclosure docs, and live demo harness.

THANK YOU



PROOFLINE

Evidence-Backed AI for Brand Governance

Every claim checked. Every finding traceable.

Live Demo Link

GitHub Repository

Team / Contact
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